From Data to Glory in the NCAA Men’s Basketball Tournament

Ben Schwartz, Matthew Shrake
Top Tier Bracketologists

Team Members

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Incoming Financial Analyst Intern
Leader Bank

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Incoming Baseball R&D Intern
Independent Sports & Entertainment

Sports Analytics Club Presidents

Prior Research Project:
“Determination of Winning Baseball Strategies Under MLB’s Extra-Inning Rule Modifications”

Agenda

1. Data Cleaning & Processing

2. Seed-Based Analysis

3. Feature Engineering & Selection

4. Model Selection & Evaluation

5. 2026 March Madness Predictions

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Data Cleaning & Processing

Data Inventory

- 38 files from Kaggle narrowed down to 7 files for analysis
- Methods
  - Dataset audit (missing/duplicate values)
  - Domain expertise (College Basketball knowledge)

Data Cleaning and Processing

Key File: Tournament Matchups

Team-Level Datasets

- KenPom/BartTorvik
- BartTorvik Away-Neutral
- EvanMiya
- RPPF Ratings
- Resumes
- Shooting Splits

Key Challenges

- Inconsistent team naming across datasets
- Missing values across sources
- Multiple formats and column naming conventions
- Need to align team level and game-level data

Data Cleaning & Processing

Cleaning Steps

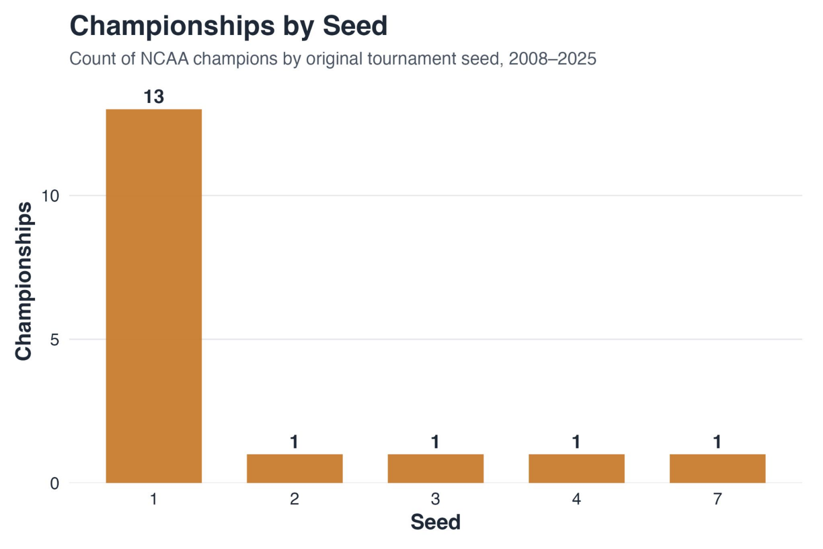
- Standardized team names
  - Lowercasing, removing punctuation, mapping aliases (e.g., “UConn” → “connecticut”)
- Aligned keys
  - Created consistent (year, team) identifiers across all datasets
- Handled duplicates
  - Ensured one row per team per season
- Validated merges
  - Checked missingness and team coverage by year

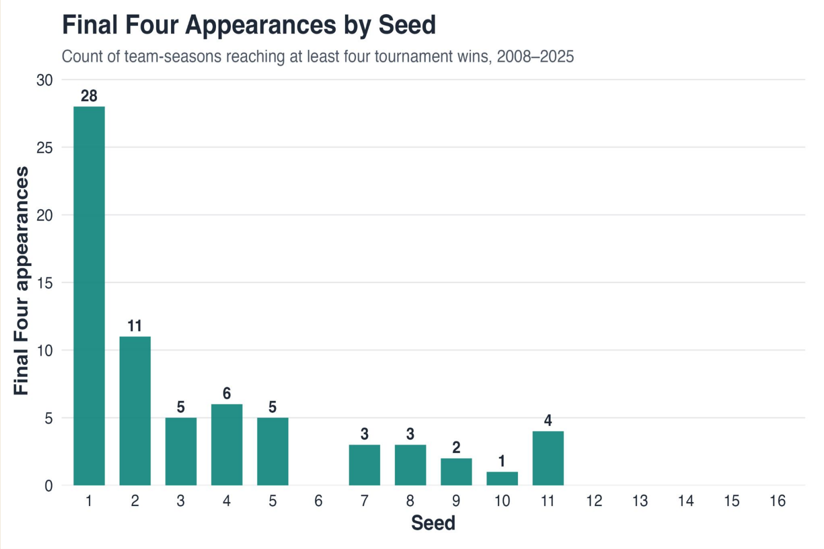
Data Quality Checks

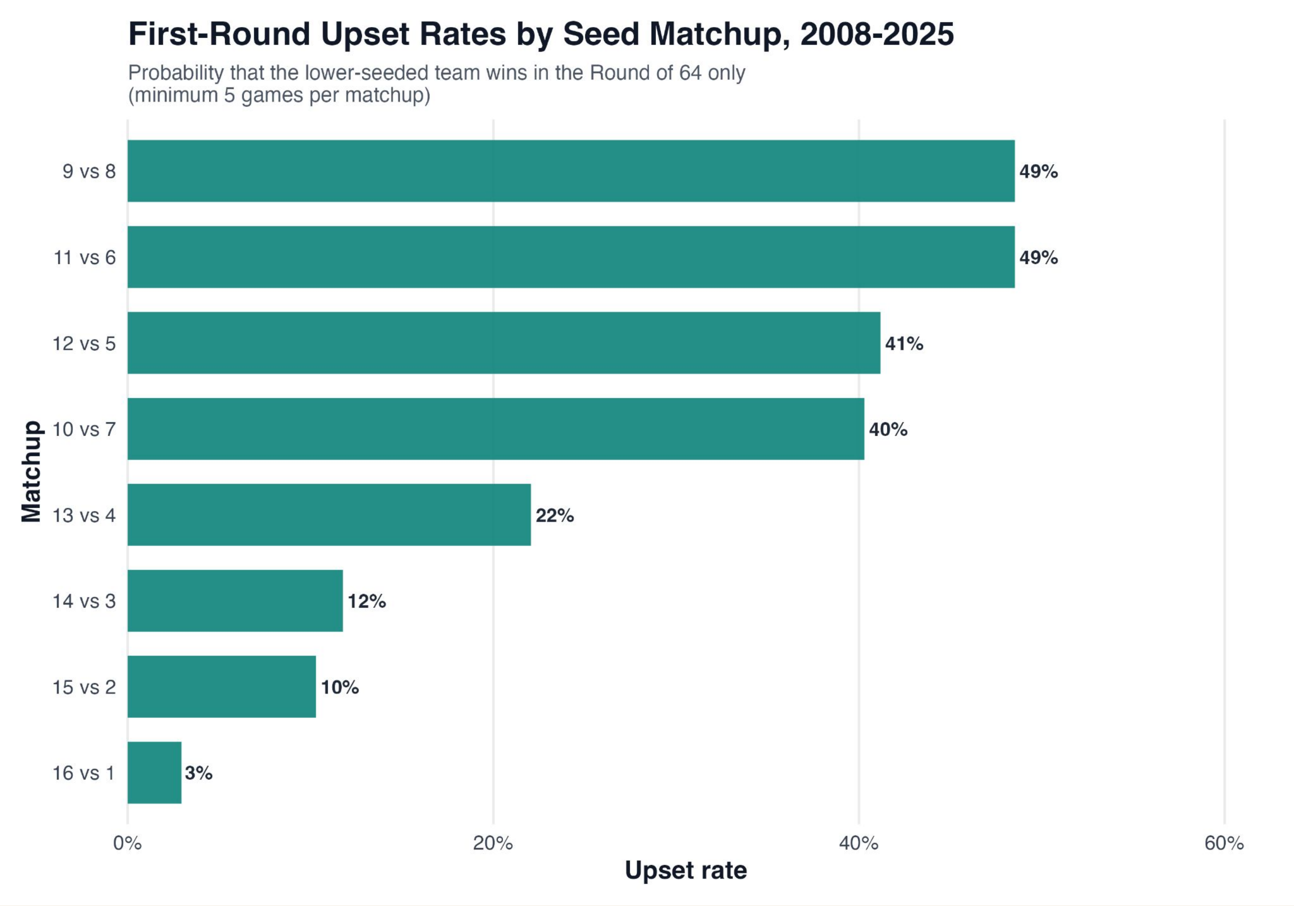
- Verified:
  - ~68 teams per year (tournament teams)
  - No duplicate (year, team) rows
  - Consistent merge across datasets

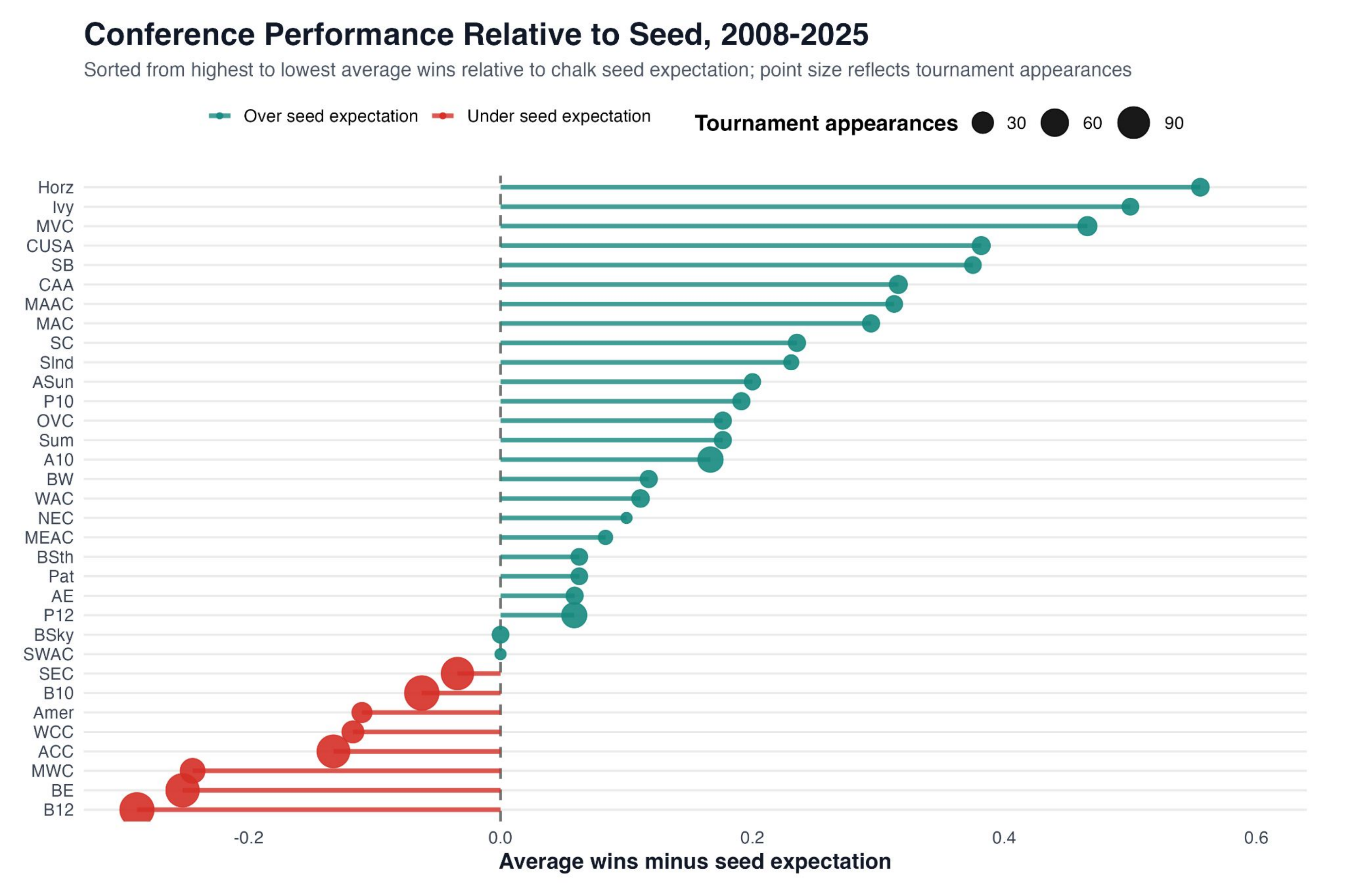
2

Seed-Based Analysis


Championships by Seed
Count of NCAA champions by original tournament seed, 2008–2025
Championships
Seed
13
1
1
1
1
0
5
10
1
2
3
4
7


Final Four Appearances by Seed
Count of team-seasons reaching at least four tournament wins, 2008–2025
Final Four appearances
Seed
28
11
5
6
5
3
3
2
1
4
0
5
10
15
20
25
30
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16


First-Round Upset Rates by Seed Matchup, 2008-2025
Probability that the lower-seeded team wins in the Round of 64 only
(minimum 5 games per matchup)
Matchup
9 vs 8
11 vs 6
12 vs 5
10 vs 7
13 vs 4
14 vs 3
15 vs 2
16 vs 1
49%
49%
41%
40%
22%
12%
10%
3%
0%
20%
40%
60%
Upset rate



Conference Performance Relative to Seed, 2008-2025
Sorted from highest to lowest average wins relative to chalk seed expectation; point size reflects tournament appearances
Over seed expectation
Under seed expectation
Tournament appearances
30
60
90
Horz
Ivy
MVC
CUSA
SB
CAA
MAAC
MAC
SC
SInd
ASun
P10
OVC
Sum
A10
BW
WAC
NEC
MEAC
BSth
Pat
AE
P12
BSky
SWAC
SEC
B10
Amer
WCC
ACC
MWC
BE
B12
-0.2
0.0
0.2
0.4
0.6
Average wins minus seed expectation

Understanding Actual Wins vs. Expected Wins

Actual Wins

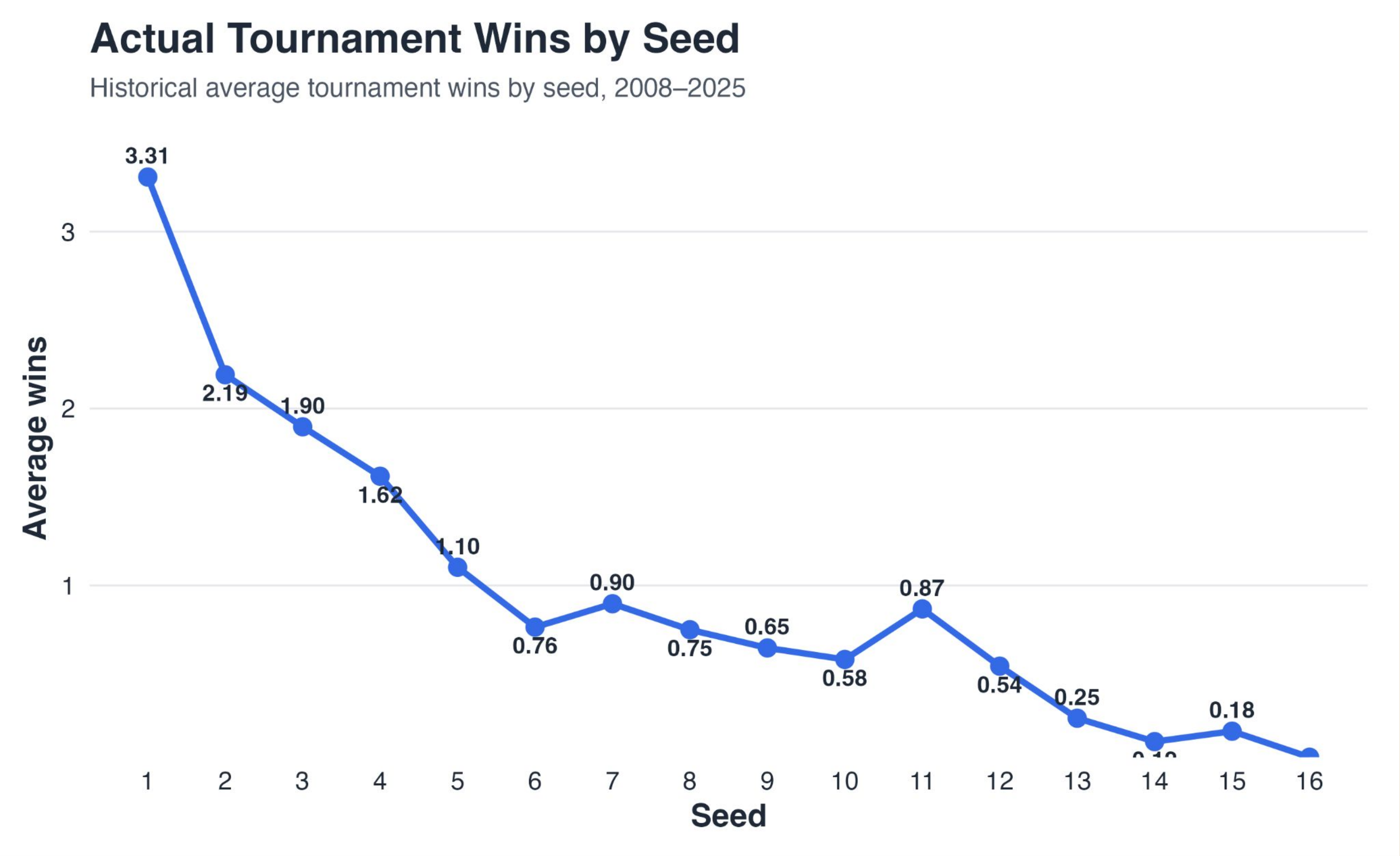
For the purposes of this project, “actual wins” refer to historical totals.

- Helpful to understand the context of the tournament before attempting to predict 2026 events

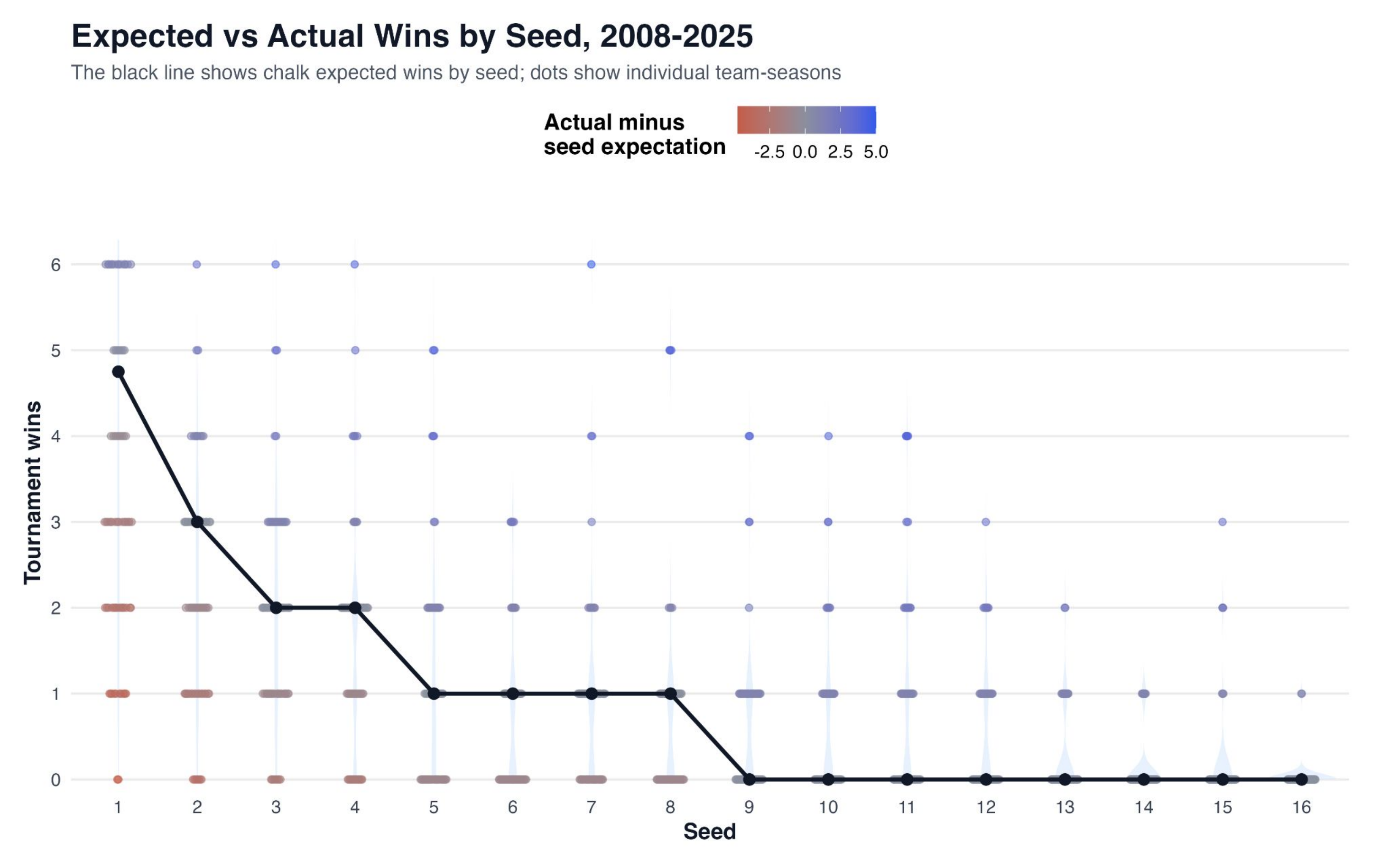
Expected Wins

Based on the assumption that the better seed will win in each round; i.e. a chalk bracket.

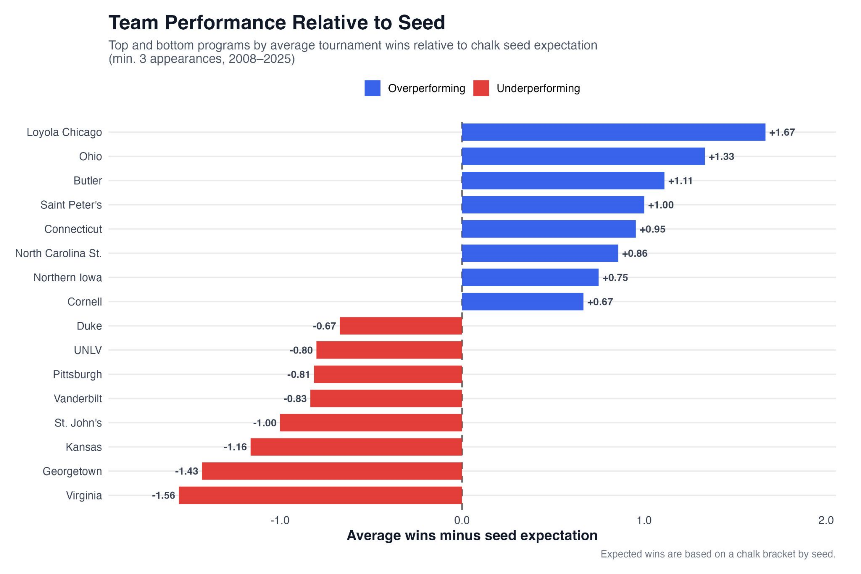
- #9-16: 0 wins
- #5-8 seeds: 1 win
- #3-4 seeds: 2 wins
- #2 seeds: 3 wins
- #1 seeds: 4.75 wins
    - #1 seeds win 19 games combined under this chalk bracket logic



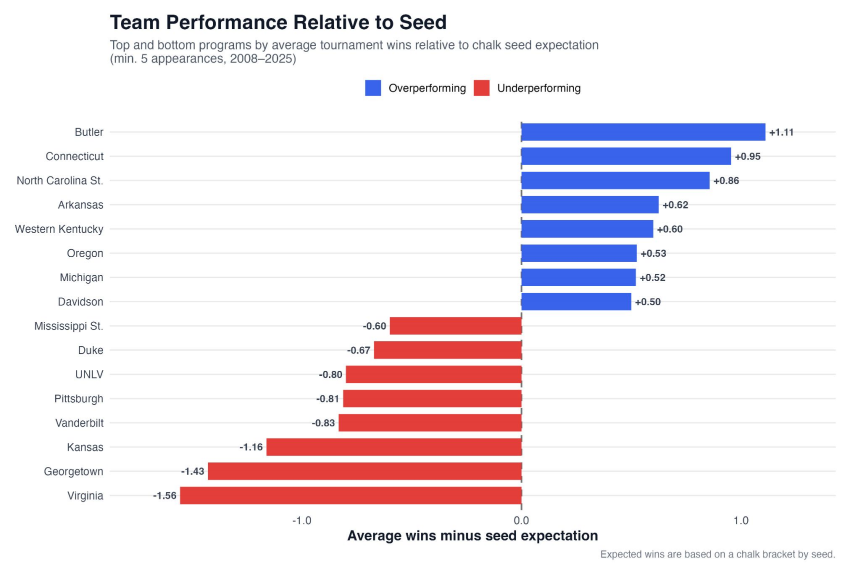
Actual Tournament Wins by Seed
Historical average tournament wins by seed, 2008–2025
Average wins
Seed
3.31
2.19
1.90
1.62
1.10
0.76
0.90
0.75
0.65
0.58
0.87
0.54
0.25
0.18
3
2
1
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16



Expected vs Actual Wins by Seed, 2008-2025
The black line shows chalk expected wins by seed; dots show individual team-seasons
Actual minus seed expectation
-2.5 0.0 2.5 5.0
Tournament wins
0
1
2
3
4
5
6
Seed
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



Team Performance Relative to Seed
Top and bottom programs by average tournament wins relative to chalk seed expectation
(min. 3 appearances, 2008–2025)
Overperforming
Underperforming
Loyola Chicago
Ohio
Butler
Saint Peter's
Connecticut
North Carolina St.
Northern Iowa
Cornell
Duke
UNLV
Pittsburgh
Vanderbilt
St. John's
Kansas
Georgetown
Virginia
+1.67
+1.33
+1.11
+1.00
+0.95
+0.86
+0.75
+0.67
-0.67
-0.80
-0.81
-0.83
-1.00
-1.16
-1.43
-1.56
-1.0
0.0
1.0
2.0
Average wins minus seed expectation
Expected wins are based on a chalk bracket by seed.



Team Performance Relative to Seed
Top and bottom programs by average tournament wins relative to chalk seed expectation
(min. 5 appearances, 2008–2025)
Overperforming
Underperforming
Butler
Connecticut
North Carolina St.
Arkansas
Western Kentucky
Oregon
Michigan
Davidson
Mississippi St.
Duke
UNLV
Pittsburgh
Vanderbilt
Kansas
Georgetown
Virginia
+1.11
+0.95
+0.86
+0.62
+0.60
+0.53
+0.52
+0.50
-0.60
-0.67
-0.80
-0.81
-0.83
-1.16
-1.43
-1.56
-1.0
0.0
1.0
Average wins minus seed expectation
Expected wins are based on a chalk bracket by seed.

3

Feature Engineering & Selection

Feature Engineering

Team-Level → Game-Level

- Merged team features onto each game:
  - Categorized into Team 1 features and Team 2 features

Difference Features

- Difference for a feature = Team 1 value - Team 2 value
- Examples:
  - kp_adj_em_diff
  - bart_an_adj_d_diff
  - Threes_fg_pct_diff
- Captures relative team strength, which drives outcomes

Data Quality Improvements

Preventing Data Leakage

- Removed all post-game variables
  - Scores
  - Winners
  - Round reached
  - Upset indicators
- Ensured model only uses pre-game information

Handling Missing Data

- Conducted missingness analysis
  - Identified high-missing features (>60%)
- Created missing indicators for partial coverage features
- Imputed remaining values using column means

Feature Selection

- Retained:
  - Efficiency metrics (KenPom, Barttorvik)
  - Resume / strength of schedule metrics
  - Shooting and rebounding statistics
- Removed:
  - Highly sparse features (e.g., RPPF, some EvanMiya)
  - Redundant or low-signal variables

Final Dataset

- 1,070 games
- 48 engineered features
- Clean, leakage-free modeling dataset

4

Model Selection & Evaluation

Model Comparisons

	LOGISTIC REGRESSION	RANDOM FOREST	GRADIENT BOOSTING
INTERPRETABILITY	High (clear coefficients)	Low	Very Low
TRANSPARENCY	Fully Explainable	Limited	Very Low
OVERFITTING RISK	Low	Moderate	Higher
STABILITY ACROSS SEASONS	High	Moderate	Moderate
CAPTURES NONLINEAR PATTERNS	Limited	Yes	Yes
ACCURACY (TRAIN | TEST)	71.39% | 72.62 %	86.43% | 72.62 %	91.69% | 70.24%
CONSISTENCY OF FEATURES	Strong	Strong	Strong

Defense Wins Championships

High Importance Model Features:

Features are shown with their importance rank (out of 48 total features)

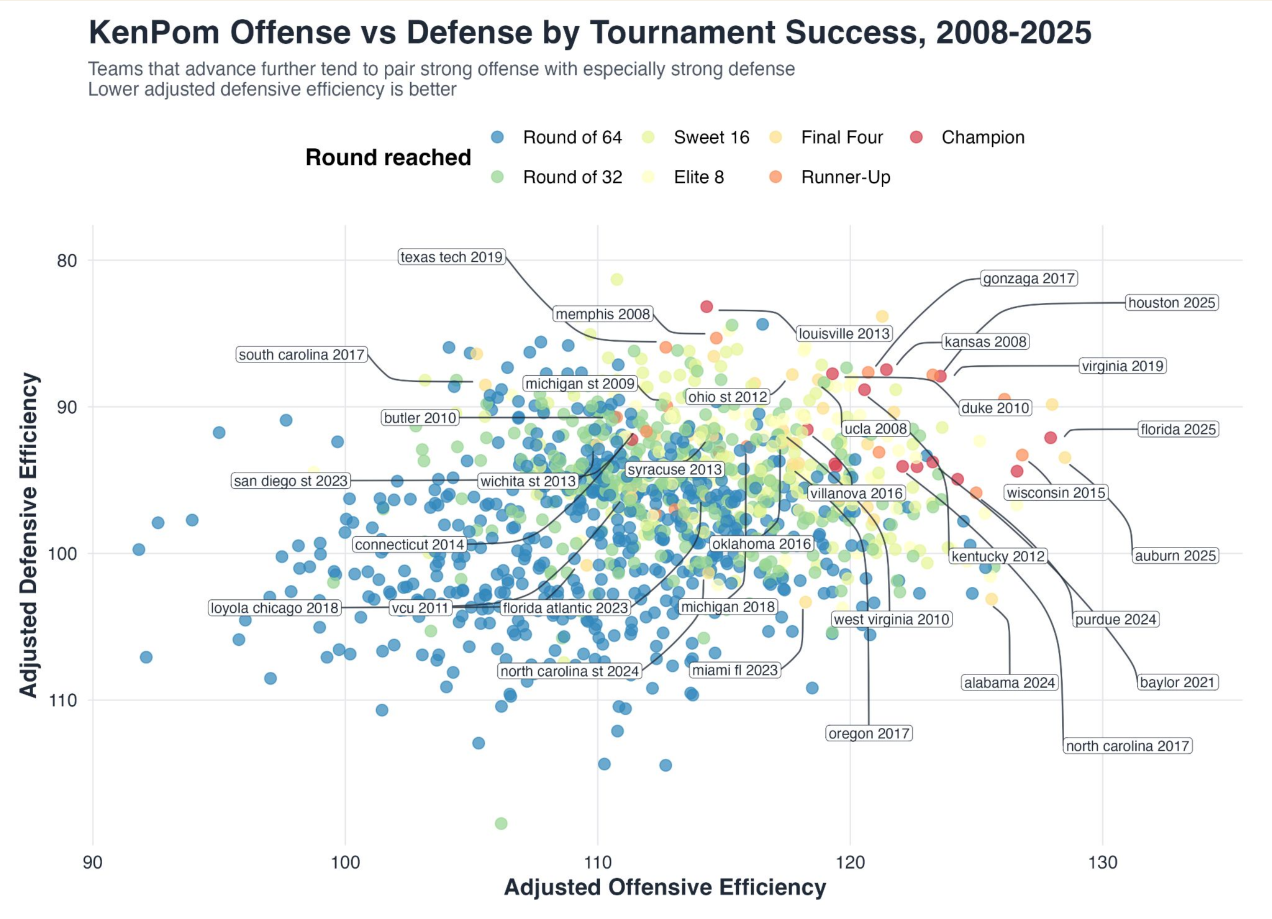
- KenPom Adjusted Defense Difference (1/48)
- Barttorvik Away-Neutral Adjusted Defense Difference (7/48)

Offense, Well-Roundedness Important Too

High Importance Model Features:

Features are shown with their importance rank (out of 48 total features)

- KenPom Adjusted Efficiency Margin Difference (4/48)
- KenPom-Barttorvik Aggregate Efficiency Margin Difference (9/48)



KenPom Offense vs Defense by Tournament Success, 2008-2025
Teams that advance further tend to pair strong offense with especially strong defense
Lower adjusted defensive efficiency is better
Round reached
Round of 64
Round of 32
Sweet 16
Elite 8
Final Four
Runner-Up
Champion
Adjusted Defensive Efficiency
Adjusted Offensive Efficiency
80
90
100
110
90
100
110
120
130
texas tech 2019
memphis 2008
louisville 2013
gonzaga 2017
houston 2025
kansas 2008
virginia 2019
south carolina 2017
michigan st 2009
ohio st 2012
duke 2010
butler 2010
ucla 2008
florida 2025
syracuse 2013
san diego st 2023
wichita st 2013
villanova 2016
wisconsin 2015
connecticut 2014
oklahoma 2016
kentucky 2012
auburn 2025
loyola chicago 2018
vcu 2011
florida atlantic 2023
michigan 2018
west virginia 2010
purdue 2024
north carolina st 2024
miami fl 2023
alabama 2024
baylor 2021
oregon 2017
north carolina 2017

Who Did You Play, How Did You Do?

(Strength of Schedule & Resume)

High Importance Model Features:

Features are shown with their importance rank (out of 48 total features)

- Resume Elo Difference (2/48)
- Resume B Power Difference (3/48)
- Resume Wins Above Bubble Rank Difference (5/48)
- KenPom Wins Above Bubble Difference (8/48)

Good Teams Shoot Well & Get Second Chances

High Importance Model Features:

Features are shown with their importance rank (out of 48 total features)

- Three-Point Field Goal Percentage Difference (4/48)
- KenPom Offensive Rebounding Difference (9/48)

Model Limitations

- No Explicit Momentum/Time Dynamics
  - May miss “hot” or “cold” teams heading into the tournament
- Potential Overconfidence (Chalk Bias)
  - Bracket predictions may be too conservative
- No Injury or Roster Information
  - Can mis-evaluate teams late in the season
- No In-Tournament Updates (Static Model)
  - Cannot adapt to new information during the tournament

5

2026 March Madness Predictions

Prediction Methods

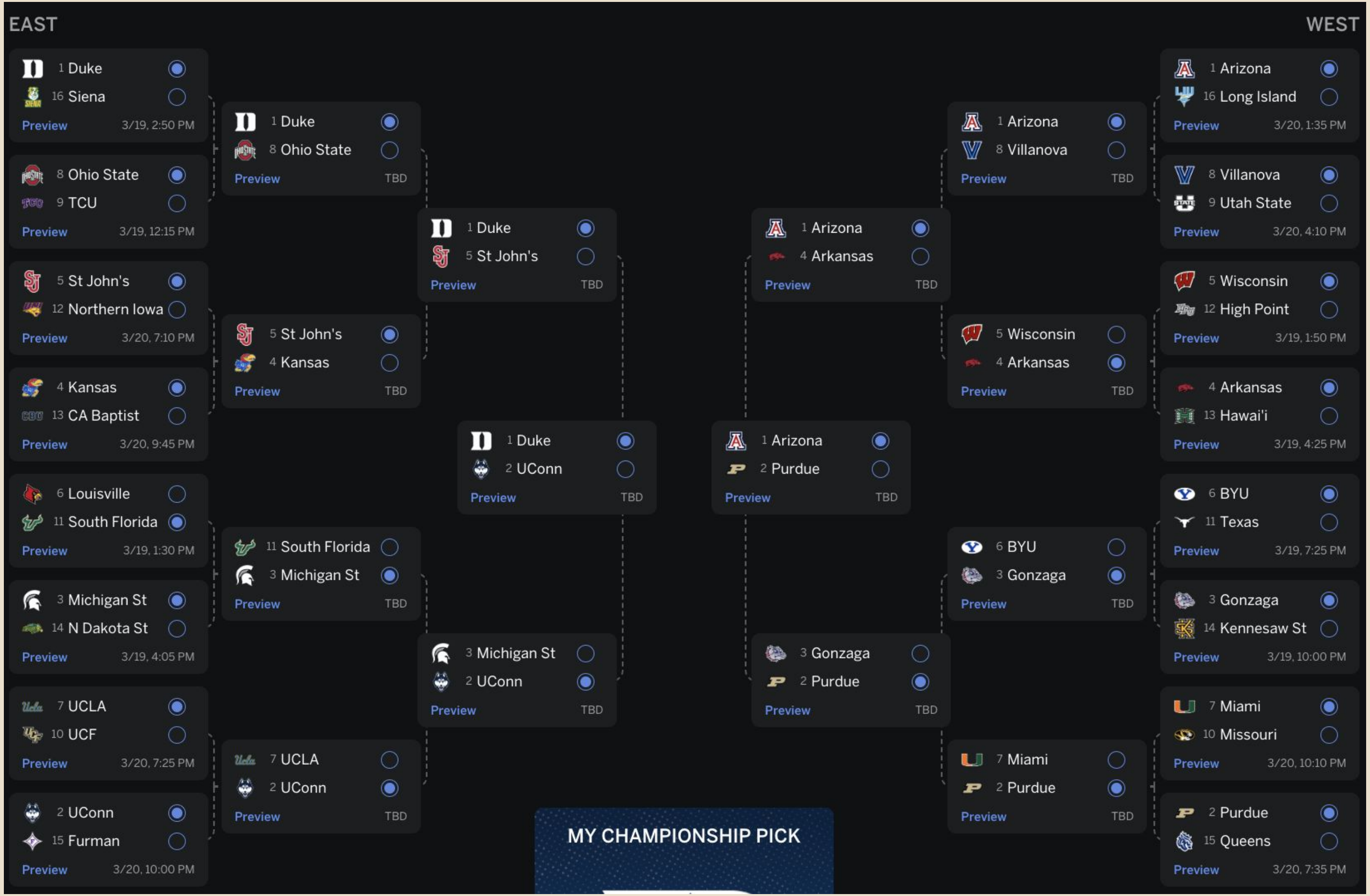
Deterministic

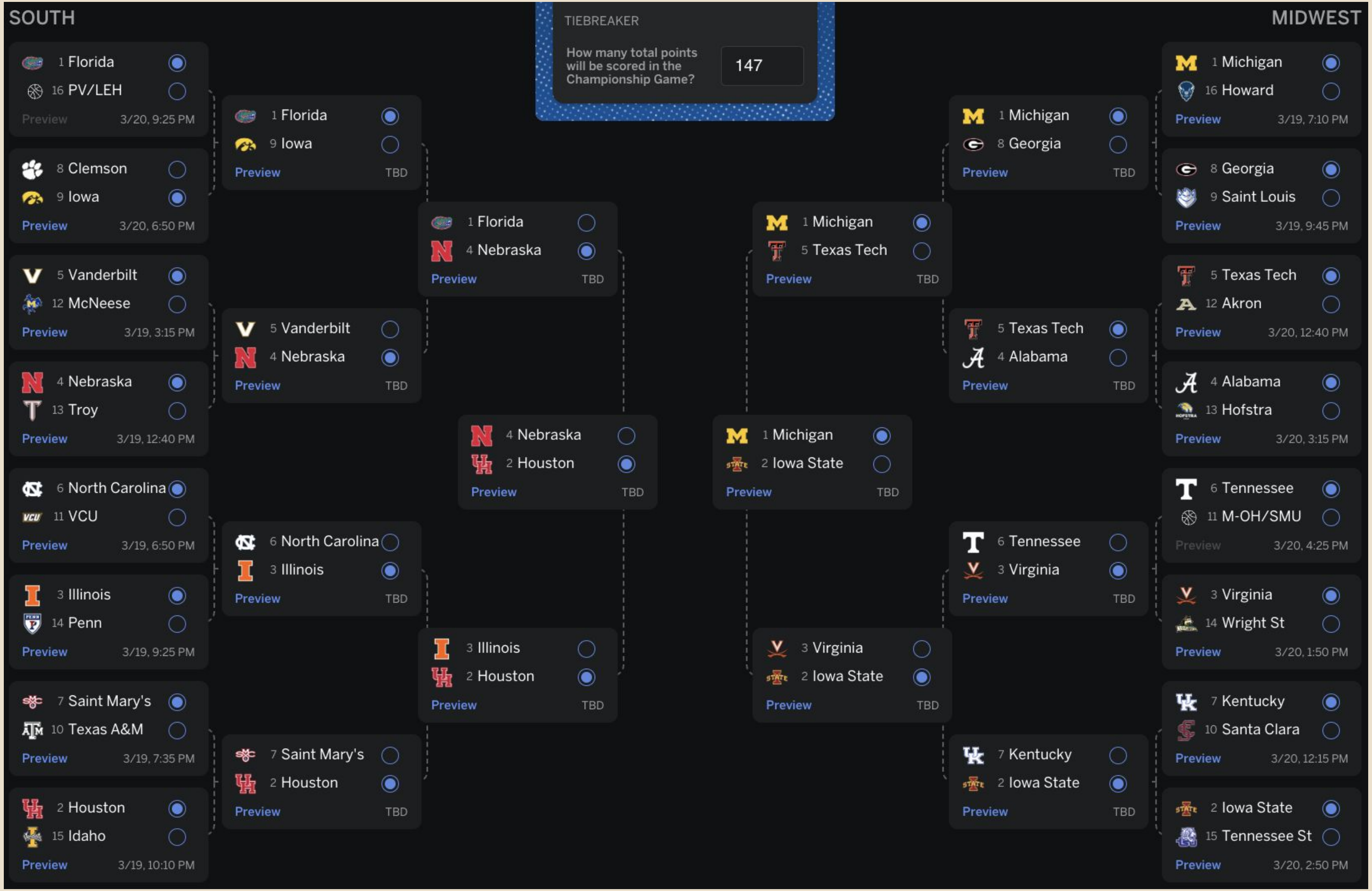
- “Team 1” vs. “Team 2”
- Logistic Regression model predicts probability of “team_1_win”
- Winning team advanced in the bracket
- Output: One bracket created using these raw probabilities

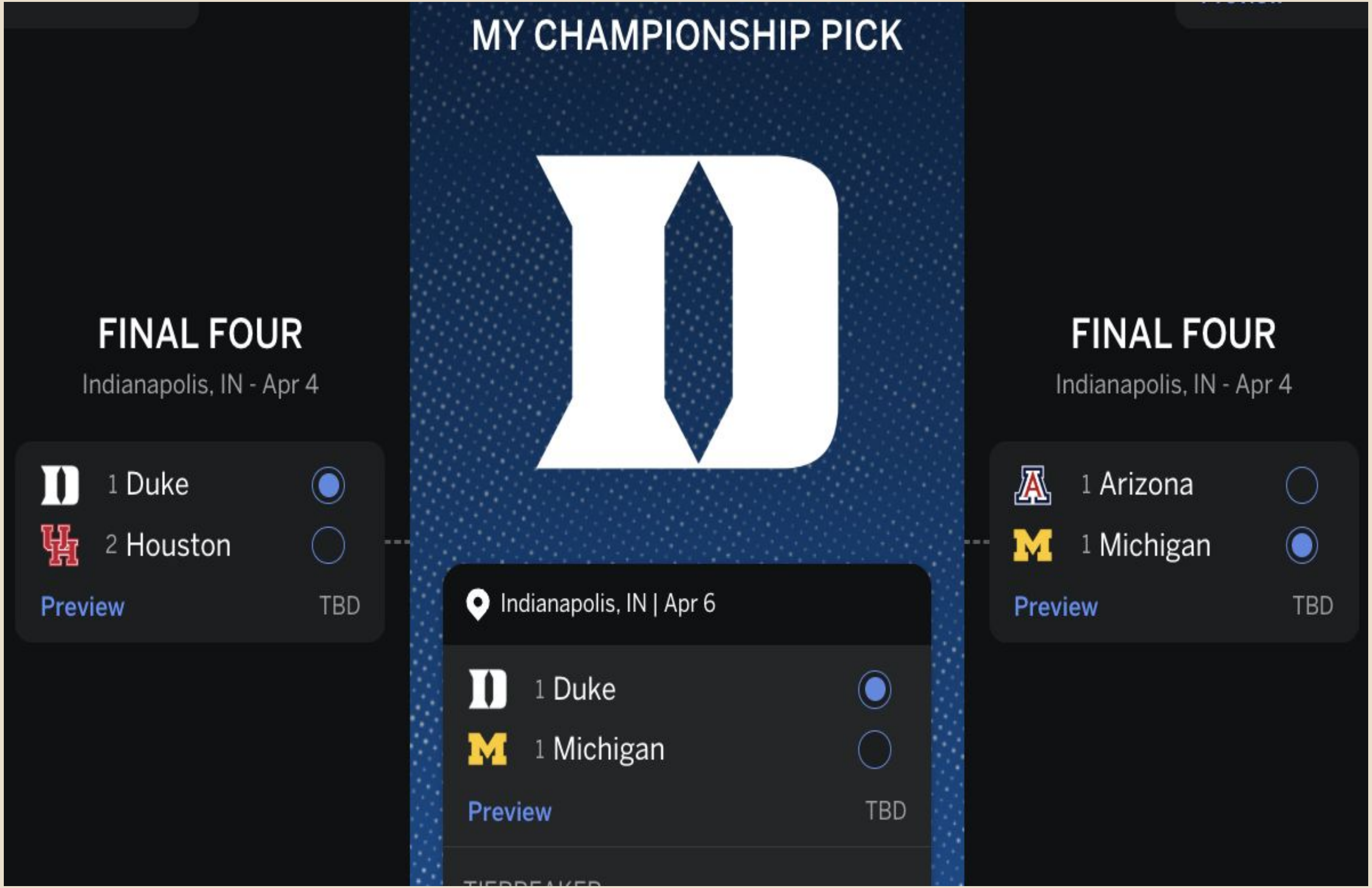
Monte Carlo

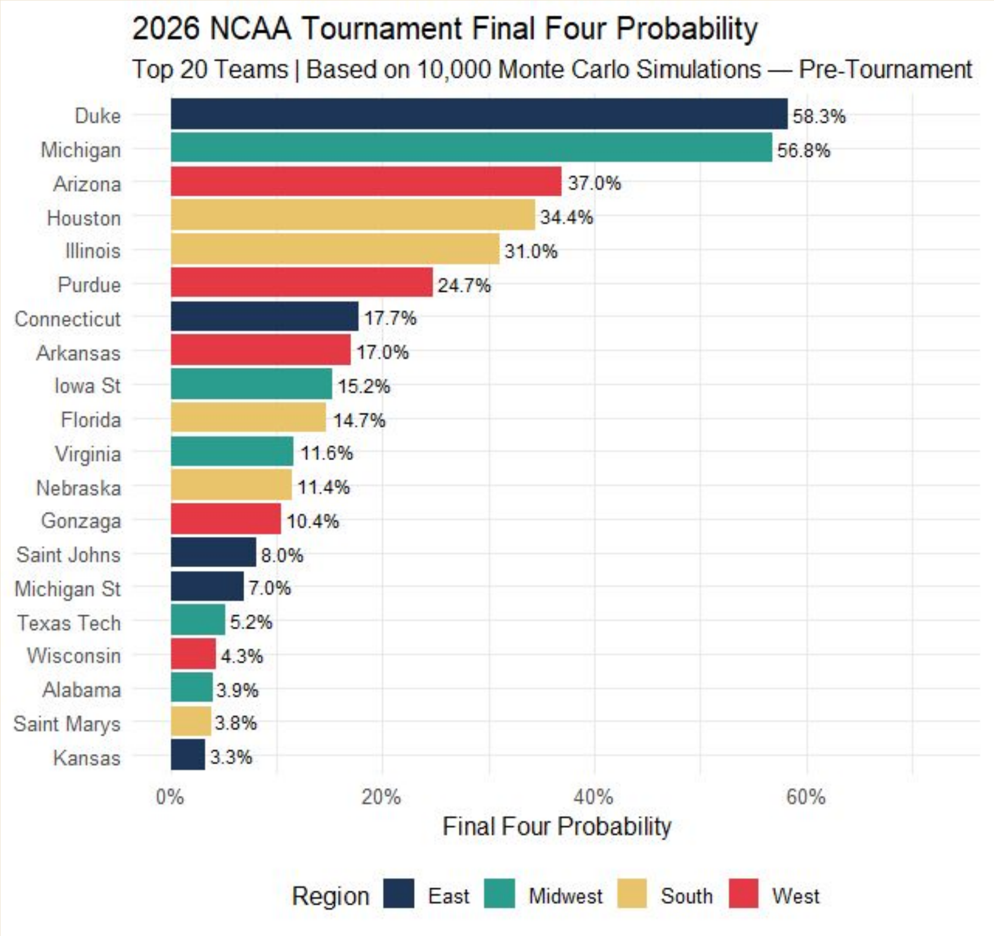
- Tournament bracket simulated many (10,000) times using logistic regression raw probabilities
- Output: Probabilities of each team reaching individual rounds of the bracket (i.e., Sweet 16, Final 4)

Pre-Tournament Predictions

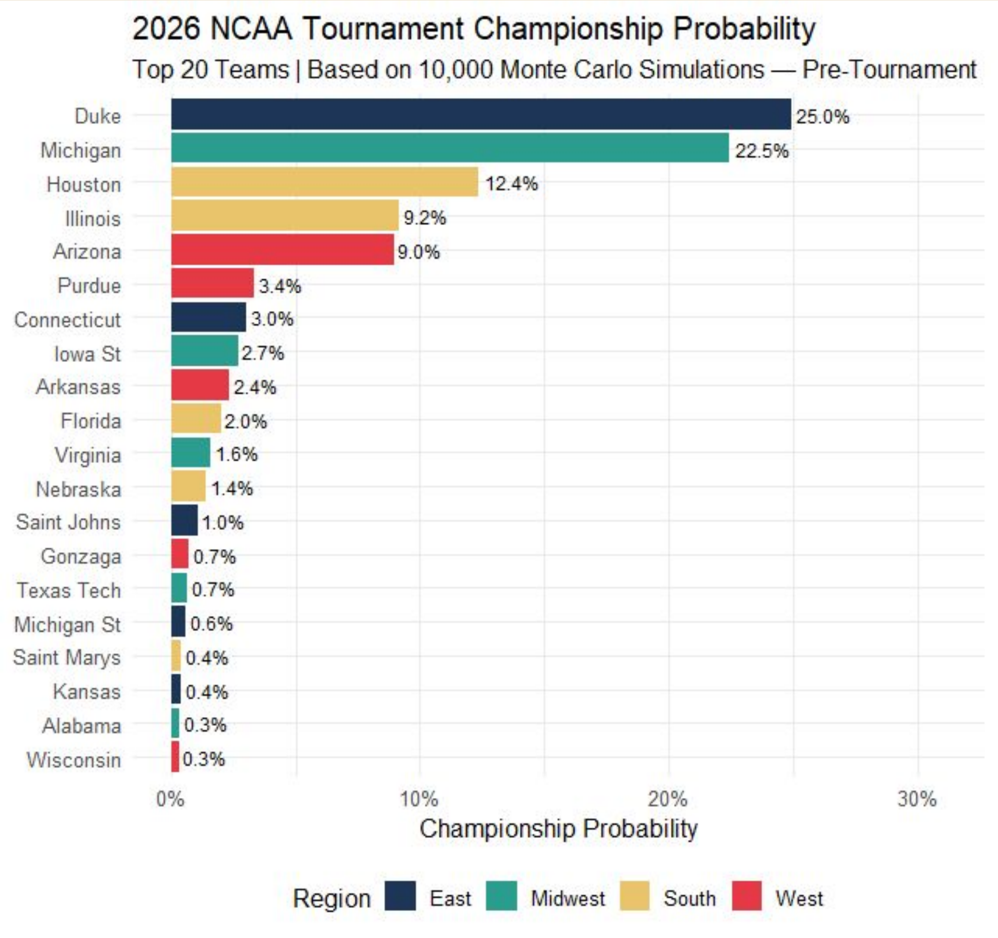

EAST
WEST
1 Duke
16 Siena
Preview 3/19, 2:50 PM
8 Ohio State
9 TCU
Preview 3/19, 12:15 PM
5 St John's
12 Northern Iowa
Preview 3/20, 7:10 PM
4 Kansas
13 CA Baptist
Preview 3/20, 9:45 PM
6 Louisville
11 South Florida
Preview 3/19, 1:30 PM
3 Michigan St
14 N Dakota St
Preview 3/19, 4:05 PM
7 UCLA
10 UCF
Preview 3/20, 7:25 PM
2 UConn
15 Furman
Preview 3/20, 10:00 PM
1 Duke
8 Ohio State
Preview TBD
5 St John's
4 Kansas
Preview TBD
11 South Florida
3 Michigan St
Preview TBD
7 UCLA
2 UConn
Preview TBD
1 Duke
5 St John's
Preview TBD
3 Michigan St
2 UConn
Preview TBD
1 Duke
2 UConn
Preview TBD
1 Arizona
2 Purdue
Preview TBD
1 Arizona
4 Arkansas
Preview TBD
3 Gonzaga
2 Purdue
Preview TBD
1 Arizona
8 Villanova
Preview TBD
5 Wisconsin
4 Arkansas
Preview TBD
6 BYU
3 Gonzaga
Preview TBD
7 Miami
2 Purdue
Preview TBD
1 Arizona
16 Long Island
Preview 3/20, 1:35 PM
8 Villanova
9 Utah State
Preview 3/20, 4:10 PM
5 Wisconsin
12 High Point
Preview 3/19, 1:50 PM
4 Arkansas
13 Hawai'i
Preview 3/19, 4:25 PM
6 BYU
11 Texas
Preview 3/19, 7:25 PM
3 Gonzaga
14 Kennesaw St
Preview 3/19, 10:00 PM
7 Miami
10 Missouri
Preview 3/20, 10:10 PM
2 Purdue
15 Queens
Preview 3/20, 7:35 PM
MY CHAMPIONSHIP PICK


SOUTH
MIDWEST
TIEBREAKER
How many total points will be scored in the Championship Game?
147
1 Florida
16 PV/LEH
Preview 3/20, 9:25 PM
8 Clemson
9 Iowa
Preview 3/20, 6:50 PM
5 Vanderbilt
12 McNeese
Preview 3/19, 3:15 PM
4 Nebraska
13 Troy
Preview 3/19, 12:40 PM
6 North Carolina
11 VCU
Preview 3/19, 6:50 PM
3 Illinois
14 Penn
Preview 3/19, 9:25 PM
7 Saint Mary's
10 Texas A&M
Preview 3/19, 7:35 PM
2 Houston
15 Idaho
Preview 3/19, 10:10 PM
1 Florida
9 Iowa
Preview TBD
5 Vanderbilt
4 Nebraska
Preview TBD
6 North Carolina
3 Illinois
Preview TBD
7 Saint Mary's
2 Houston
Preview TBD
1 Florida
4 Nebraska
Preview TBD
3 Illinois
2 Houston
Preview TBD
4 Nebraska
2 Houston
Preview TBD
1 Michigan
2 Iowa State
Preview TBD
1 Michigan
5 Texas Tech
Preview TBD
3 Virginia
2 Iowa State
Preview TBD
1 Michigan
8 Georgia
Preview TBD
5 Texas Tech
4 Alabama
Preview TBD
6 Tennessee
3 Virginia
Preview TBD
7 Kentucky
2 Iowa State
Preview TBD
1 Michigan
16 Howard
Preview 3/19, 7:10 PM
8 Georgia
9 Saint Louis
Preview 3/19, 9:45 PM
5 Texas Tech
12 Akron
Preview 3/20, 12:40 PM
4 Alabama
13 Hofstra
Preview 3/20, 3:15 PM
6 Tennessee
11 M-OH/SMU
Preview 3/20, 4:25 PM
3 Virginia
14 Wright St
Preview 3/20, 1:50 PM
7 Kentucky
10 Santa Clara
Preview 3/20, 12:15 PM
2 Iowa State
15 Tennessee St
Preview 3/20, 2:50 PM


MY CHAMPIONSHIP PICK
FINAL FOUR
Indianapolis, IN - Apr 4
1 Duke
2 Houston
Preview
TBD
Indianapolis, IN | Apr 6
1 Duke
1 Michigan
Preview
TBD
FINAL FOUR
Indianapolis, IN - Apr 4
1 Arizona
1 Michigan
Preview
TBD



2026 NCAA Tournament Final Four Probability
Top 20 Teams | Based on 10,000 Monte Carlo Simulations — Pre-Tournament
Duke 58.3%
Michigan 56.8%
Arizona 37.0%
Houston 34.4%
Illinois 31.0%
Purdue 24.7%
Connecticut 17.7%
Arkansas 17.0%
Iowa St 15.2%
Florida 14.7%
Virginia 11.6%
Nebraska 11.4%
Gonzaga 10.4%
Saint Johns 8.0%
Michigan St 7.0%
Texas Tech 5.2%
Wisconsin 4.3%
Alabama 3.9%
Saint Marys 3.8%
Kansas 3.3%
0%
20%
40%
60%
Final Four Probability
Region
East
Midwest
South
West


2026 NCAA Tournament Championship Probability
Top 20 Teams | Based on 10,000 Monte Carlo Simulations — Pre-Tournament
Duke
Michigan
Houston
Illinois
Arizona
Purdue
Connecticut
Iowa St
Arkansas
Florida
Virginia
Nebraska
Saint Johns
Gonzaga
Texas Tech
Michigan St
Saint Marys
Kansas
Alabama
Wisconsin
25.0%
22.5%
12.4%
9.2%
9.0%
3.4%
3.0%
2.7%
2.4%
2.0%
1.6%
1.4%
1.0%
0.7%
0.7%
0.6%
0.4%
0.4%
0.3%
0.3%
0%
10%
20%
30%
Championship Probability
Region
East
Midwest
South
West

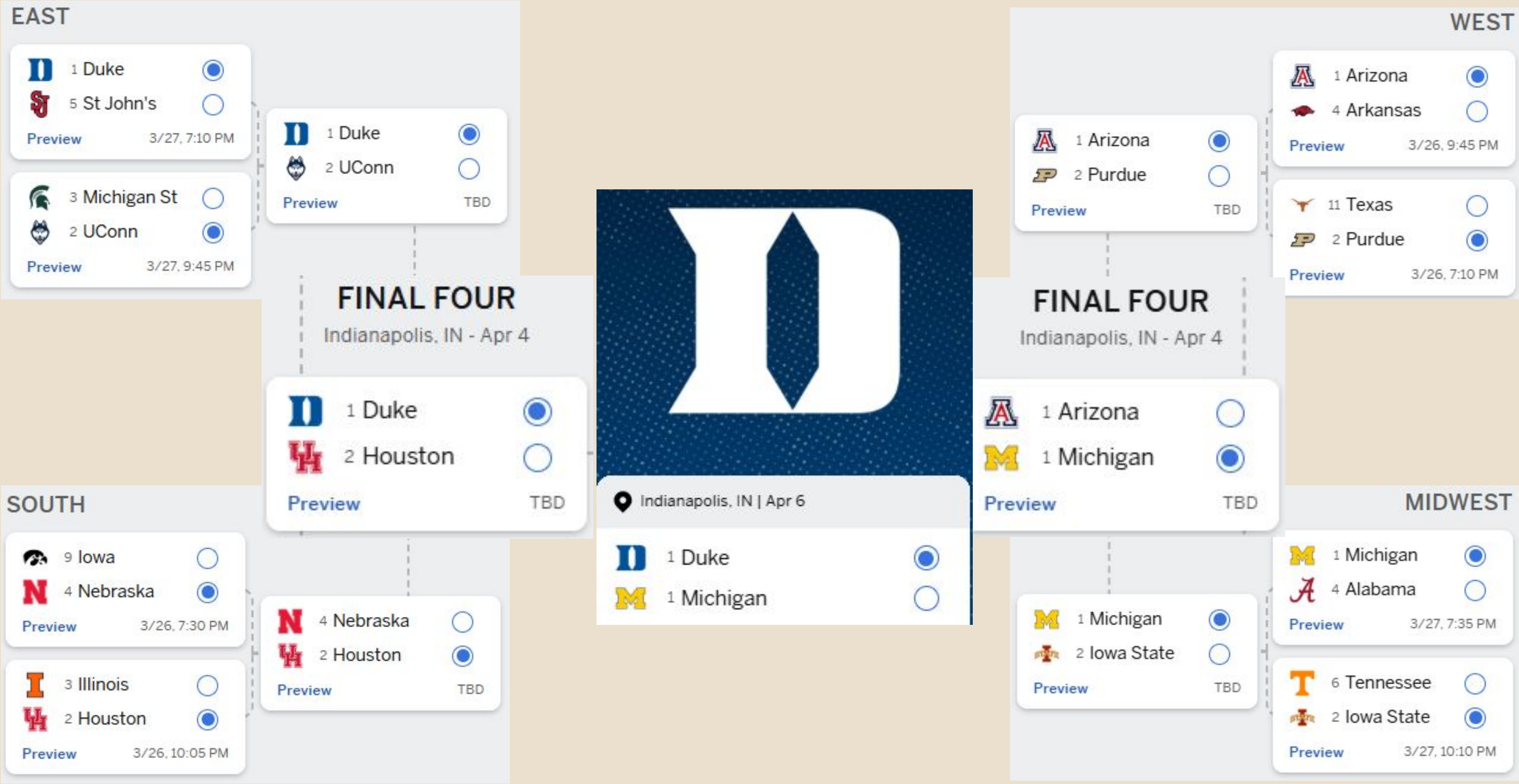
2026 Tournament Advancement Probability by Round

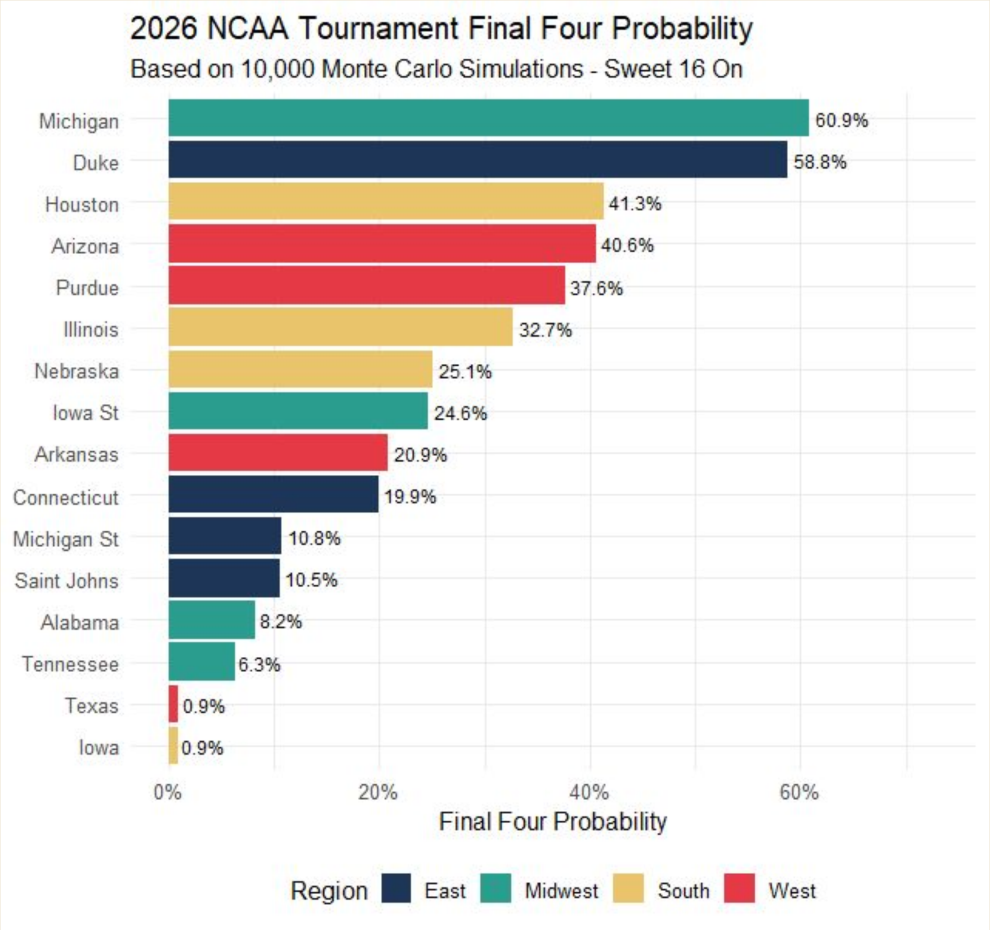
Top 20 Teams | Based on 10,000 Monte Carlo Simulations — Pre-Tournament

	Sweet 16	Elite 8	Final Four	Title Game	Champion
Duke	92.5%	74.1%	58.3%	37.8%	25.0%
Michigan	90.0%	74.6%	56.8%	40.2%	22.5%
Houston	76.5%	46.2%	34.4%	19.8%	12.4%
Illinois	85.0%	43.9%	31.0%	16.7%	9.2%
Arizona	79.8%	53.2%	37.0%	18.7%	9.0%
Purdue	78.0%	52.6%	24.7%	10.2%	3.4%
Connecticut	75.1%	49.2%	17.7%	7.0%	3.0%
Iowa St	67.9%	40.4%	15.2%	7.4%	2.7%
Arkansas	65.1%	29.3%	17.0%	6.7%	2.4%
Florida	85.8%	46.4%	14.7%	5.3%	2.0%
Virginia	66.5%	34.8%	11.6%	5.4%	1.6%
Nebraska	60.8%	34.4%	11.4%	3.8%	1.4%
Saint Johns	59.2%	15.7%	8.0%	2.8%	1.0%
Gonzaga	66.2%	29.2%	10.4%	2.9%	0.7%
Texas Tech	44.5%	11.1%	5.2%	2.1%	0.7%
Michigan St	55.3%	25.5%	7.0%	2.2%	0.6%
Saint Marys	22.2%	7.3%	3.8%	1.2%	0.4%
Kansas	38.5%	8.3%	3.3%	1.0%	0.4%
Alabama	40.7%	9.2%	3.9%	1.5%	0.3%
Wisconsin	29.8%	9.7%	4.3%	1.3%	0.3%

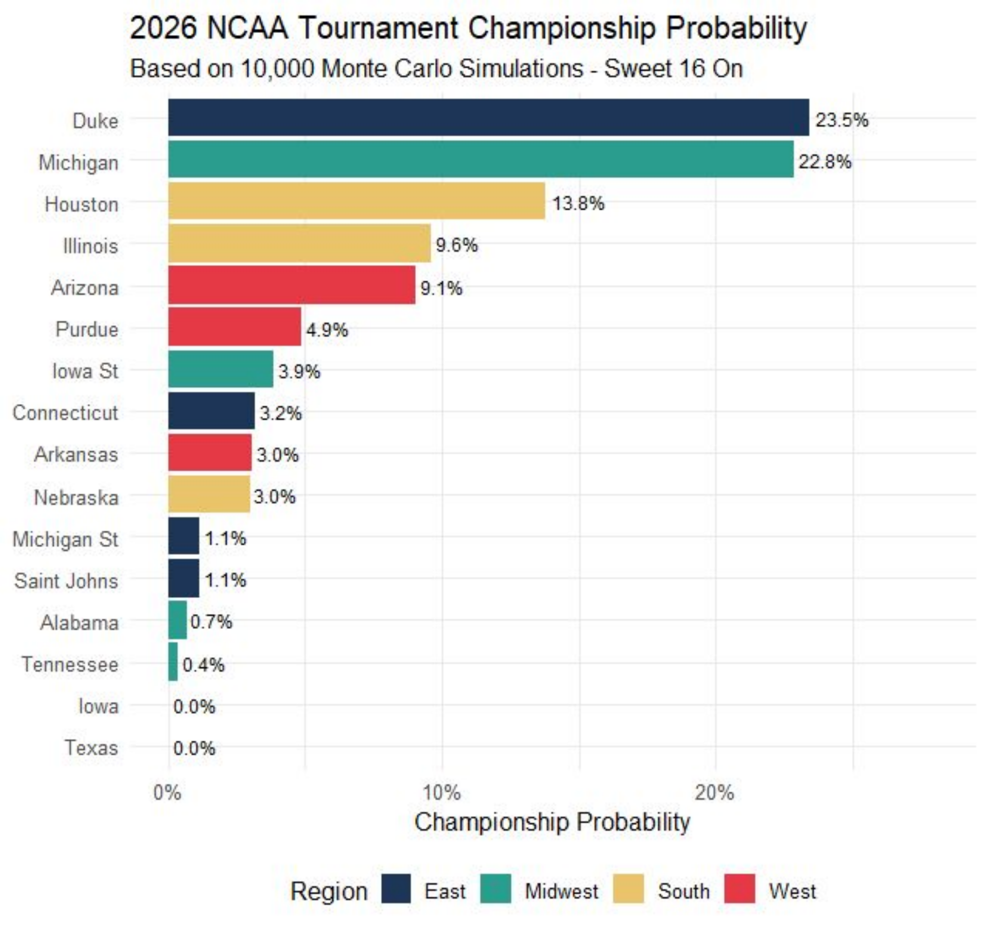
Probability 25% 50% 75%

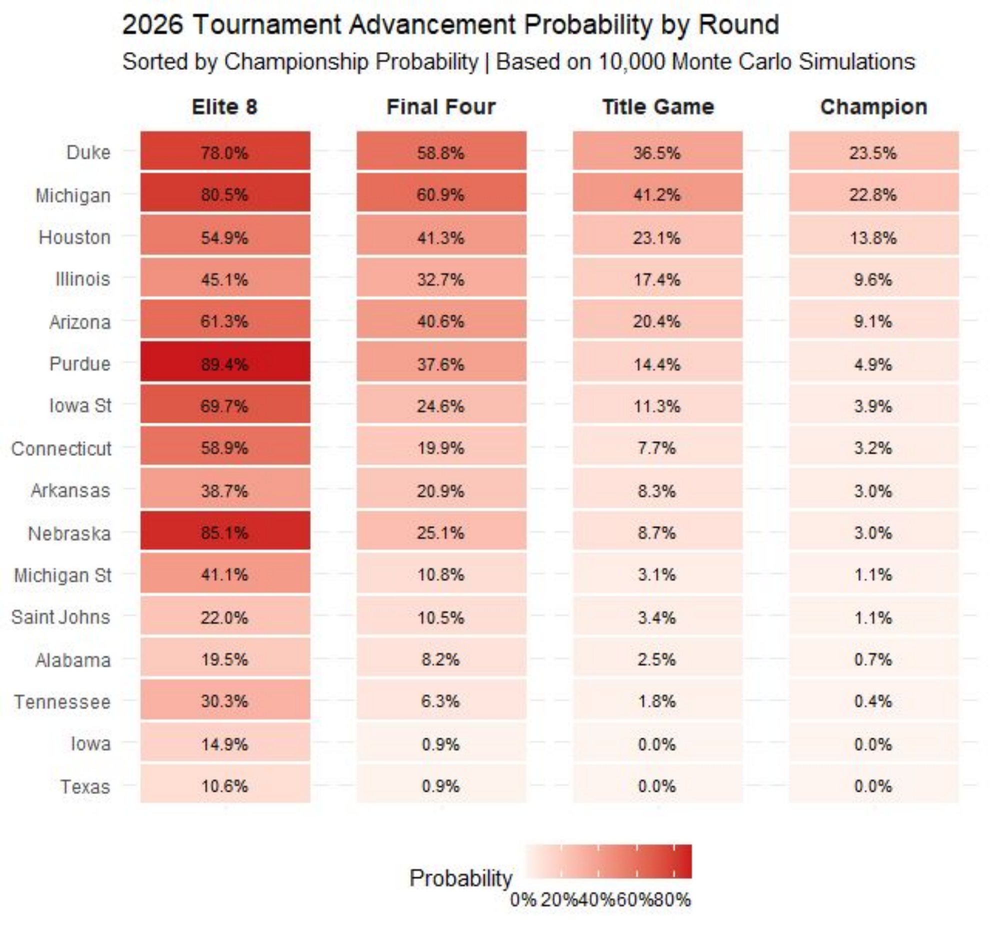
Pre-Sweet 16 Predictions


EAST
1 Duke
5 St John's
Preview 3/27, 7:10 PM
3 Michigan St
2 UConn
Preview 3/27, 9:45 PM
1 Duke
2 UConn
Preview TBD
FINAL FOUR
Indianapolis, IN - Apr 4
1 Duke
2 Houston
Preview TBD
SOUTH
9 Iowa
4 Nebraska
Preview 3/26, 7:30 PM
3 Illinois
2 Houston
Preview 3/26, 10:05 PM
4 Nebraska
2 Houston
Preview TBD
Indianapolis, IN | Apr 6
1 Duke
1 Michigan
WEST
1 Arizona
4 Arkansas
Preview 3/26, 9:45 PM
11 Texas
2 Purdue
Preview 3/26, 7:10 PM
1 Arizona
2 Purdue
Preview TBD
FINAL FOUR
Indianapolis, IN - Apr 4
1 Arizona
1 Michigan
Preview TBD
MIDWEST
1 Michigan
4 Alabama
Preview 3/27, 7:35 PM
6 Tennessee
2 Iowa State
Preview 3/27, 10:10 PM
1 Michigan
2 Iowa State
Preview TBD



2026 NCAA Tournament Final Four Probability
Based on 10,000 Monte Carlo Simulations - Sweet 16 On
Michigan
Duke
Houston
Arizona
Purdue
Illinois
Nebraska
Iowa St
Arkansas
Connecticut
Michigan St
Saint Johns
Alabama
Tennessee
Texas
Iowa
60.9%
58.8%
41.3%
40.6%
37.6%
32.7%
25.1%
24.6%
20.9%
19.9%
10.8%
10.5%
8.2%
6.3%
0.9%
0.9%
0%
20%
40%
60%
Final Four Probability
Region
East
Midwest
South
West


2026 NCAA Tournament Championship Probability
Based on 10,000 Monte Carlo Simulations - Sweet 16 On
Duke
Michigan
Houston
Illinois
Arizona
Purdue
Iowa St
Connecticut
Arkansas
Nebraska
Michigan St
Saint Johns
Alabama
Tennessee
Iowa
Texas
23.5%
22.8%
13.8%
9.6%
9.1%
4.9%
3.9%
3.2%
3.0%
3.0%
1.1%
1.1%
0.7%
0.4%
0.0%
0.0%
0%
10%
20%
Championship Probability
Region
East
Midwest
South
West


2026 Tournament Advancement Probability by Round
Sorted by Championship Probability | Based on 10,000 Monte Carlo Simulations
Elite 8
Final Four
Title Game
Champion
Duke 78.0% 58.8% 36.5% 23.5%
Michigan 80.5% 60.9% 41.2% 22.8%
Houston 54.9% 41.3% 23.1% 13.8%
Illinois 45.1% 32.7% 17.4% 9.6%
Arizona 61.3% 40.6% 20.4% 9.1%
Purdue 89.4% 37.6% 14.4% 4.9%
Iowa St 69.7% 24.6% 11.3% 3.9%
Connecticut 58.9% 19.9% 7.7% 3.2%
Arkansas 38.7% 20.9% 8.3% 3.0%
Nebraska 85.1% 25.1% 8.7% 3.0%
Michigan St 41.1% 10.8% 3.1% 1.1%
Saint Johns 22.0% 10.5% 3.4% 1.1%
Alabama 19.5% 8.2% 2.5% 0.7%
Tennessee 30.3% 6.3% 1.8% 0.4%
Iowa 14.9% 0.9% 0.0% 0.0%
Texas 10.6% 0.9% 0.0% 0.0%
Probability
0%
20%
40%
60%
80%

Resources

Amin, N. (n.d.). *March Madness data* [Data set]. Kaggle. https://www.kaggle.com/datasets/nishaanamin/march-madness-data/data

Questions?
Please reach out to
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Thank you!